

Session 6 — After-Session Checklist

ExInt II: Research Designs in SME Research · WU Vienna · SS 2026

Work through every step in order. Tick the box when done.

Reference project:

<https://github.com/vkiefner/sme-intl/tree/main>

Where to work:

- VS Code
- Terminal
- Jupyter
- GitHub

A Update README.md with new research design

VS Code

Step 1 — Copy the updated README.md into your project

The README has been updated to reflect the R&D; intensity research design. Copy it from the reference project or update yours manually:

 **README.md**

Research Question

Does R&D; intensity positively affect firm performance among European SMEs, and does firm size moderate this relationship?

Hypotheses

- H1: R&D; intensity affects RoA (expensing effect expected in short run)

- H2: Firm size positively moderates the R&D; intensity-RoA relationship

■ Note on DOI: *pifo* (foreign income) is not available in Compustat Global. *rect/sale* produces extreme outliers for SMEs. Research question updated to R&D; intensity which has 100% coverage.

VS Code

Step 2 — Update the variable table in README.md

 **README.md**

Variable	Field(s)	Formula	Role
RoA	ib, at	ib / at	Dependent (Y)
R&D; intensity	xrd, at	xrd.fillna(0) / at	Independent
R&D; x Size	-	rd_intensity x ln_at	H2 interaction
Firm size	at	log(at)	Moderator+Ctrl
Leverage	dltt, at	dltt / at	Control
CAPX intensity	capx, at	capx / at	Control
Cash ratio	che, at	che / at	Control

B Adapt 03_descriptives.py for your own project

VS Code

Step 3 — Copy the updated script from the reference project

The script has been fully updated for the R&D; intensity research design. Copy it:

 **code/03_descriptives.py**

<https://github.com/vkiefner/sme-intl/tree/main/code>

Step 4 — Check and adapt the data quality + SME filters

Find the filters section and verify they match your sample:

📄 `code/03_descriptives.py`

```
# Data quality filters

df = df[(df['at'] > 0.1) & (df['sale'] > 0) & (df['seq'] > 0)].copy()

# Remove micro-firms (negative log assets distort size plots)

df = df[df['at'] >= 1].copy() # keep firms with >= EUR 1m assets

# EU SME filter

sme_mask = (df['emp'] < 0.25) | (df['at'] <= 43)

df = df[sme_mask].copy()
```

■ *at >= 1 removes micro-firms with negative log(at) — firms under EUR 1m assets have very unstable RoA and distort the firm size plots.*

Step 5 — Adapt variable construction for your own Y and X

Find the Variable Construction section and replace with your own variables:

```
# YOUR DEPENDENT VARIABLE (Y)

df['your_dv'] = df['your_numerator'] / df['your_denominator']

# YOUR INDEPENDENT VARIABLE (X)

df['your_x'] = df['your_field'].fillna(0) / df['at']

# YOUR MODERATOR + INTERACTION

df['your_mod'] = ... # e.g. ln_at already computed

df['x_mod'] = df['your_x'] * df['your_mod']
```

■ *Keep firm size (ln_at), leverage (dltt/at), CAPX intensity, and cash ratio as controls at minimum.*

Step 6 — Update CORE_VARS, WINSORIZE_VARS, and VAR_LABELS

```
CORE_VARS = ['your_dv', 'your_x', 'ln_at', 'leverage']

WINSORIZE_VARS = ['your_dv', 'your_x', 'leverage',
                  'capx_intensity', 'cash_ratio']

VAR_LABELS = {
    'your_dv': 'Your DV label',
    'your_x': 'Your X label',
    ...
}
```

■ *Do NOT winsorize ln_at (log assets) — it is already bounded and winsorizing would distort the size distribution.*

C Run 03_descriptives.py and verify all outputs

Terminal

Step 7 — Activate venv and run the script

```
.venv\Scripts\activate
```

■ **Mac / Linux:**

```
source .venv/bin/activate
```

```
python code/03_descriptives.py
```

Expected output: sample size, winsorize ranges, R&D; firms count, then 'Saved panel_with_vars.parquet'.

VS Code

Step 8 — Verify all output files exist in the correct folders

Check that all five files were created:

File	Location	Created by
summary_statistics.csv	output/tables/	03_descriptives.py
correlation_matrix.png	output/figures/	03_descriptives.py
dv_distribution.png	output/figures/	03_descriptives.py
main_relationship.png	output/figures/	03_descriptives.py
panel_with_vars.parquet	data/processed/	03_descriptives.py

VS Code

Step 9 — Review each output file

summary_statistics.csv — check:

- R&D; intensity: median = 0 is expected (most SMEs don't report R&D;)
- RoA: mean negative, median positive is typical for listed SME panels
- All ratios (leverage, capx, cash) between 0 and 1 after winsorizing

correlation_matrix.png — check:

- R&D; vs RoA: expected negative (expensing effect)
- R&D; vs size: do larger firms invest more in R&D;?
- No pair of IVs above $|r| = 0.8$

main_relationship.png — check:

- Left: downward slope expected (expensing effect)
- Right: No R&D; firms (blue) above Has R&D; firms (red) — expected
- No vertical spike or wall of dots anywhere

■ If firm size axis shows very negative values (< -2), the ≥ 1 filter was not applied. Re-run with the filter.

D Adapt 04_regression.py for your own project

VS Code

Step 10 — Copy the updated script from the reference project

📄 **code/04_regression.py**

The script now uses `get_se()` to handle both statsmodels and linearmodels standard errors — fixing the `AttributeError` from earlier.

VS Code

Step 11 — Update DV, X_MAIN, INTERACT, and CONTROLS

code/04_regression.py

```
DV = 'your_dv'
X_MAIN = 'your_x'
INTERACT = 'x_mod' # your X * moderator
CONTROLS = ['ln_at', 'leverage', 'capx_intensity', 'cash_ratio']
```

VS Code

Step 12 — Update hypothesis comments to match your own H1 and H2

Find the H1 and H2 diagnostic print blocks at the bottom and update the interpretation text. The diagnostics print automatically — but you should add your own plain-language interpretation of the expected direction.

■ If your H1 predicts a negative effect (like the R&D; expensing effect), update the diagnostic to say H1 SUPPORTED when beta is negative and significant.

E Run 04_regression.py and interpret results

Terminal

Step 13 — Run the regression script

```
python code/04_regression.py
```

Expected output: three model results, Hausman comparison, results table, H1/H2 diagnostics.

■ If you see 'singleton groups' warning from linearmodels — firms with only 1 observation after filtering. Normal, the model still runs.

VS Code

Step 14 — Verify output file and interpret results

Check that regression_results.csv was created:

File	Location	Created by
regression_results.csv	output/tables/	04_regression.py

VS Code

Step 15 — Read and interpret the results table

Open regression_results.csv and answer these questions:

1. Is beta(X) significant in Model (2) TWFE? What sign?
2. Does the sign match your hypothesis direction?
3. Is the interaction beta(INTERACT) significant in Model (3)?
4. How large is the OLS vs FE difference?
Large difference = omitted variable bias in OLS
5. Does R2 change substantially from OLS to FE?
If yes: firm fixed effects explain important variance

Write 2-3 bullet points summarizing your results in plain language — you will use these in your research note next session.

■ A negative and significant beta on R&D; intensity is theoretically meaningful — R&D; expensing reduces current earnings. This is H1 supported, not failed.

F Run full pipeline, commit and push

Terminal

Step 16 — Run task all to confirm end-to-end pipeline

```
task all
```

All four scripts should complete without errors: pull (or skip) → clean → descriptives → regression.

■ If pull fails (WRDS timeout), run python code/02_clean.py directly since raw data already exists.

VS Code

Step 17 — Verify complete output folder

After task all, check that all outputs exist:

File	Location	Created by
summary_statistics.csv	output/tables/	03_descriptives.py
regression_results.csv	output/tables/	04_regression.py
correlation_matrix.png	output/figures/	03_descriptives.py
dv_distribution.png	output/figures/	03_descriptives.py
main_relationship.png	output/figures/	03_descriptives.py
panel_with_vars.parquet	data/processed/	03_descriptives.py
panel_clean.parquet	data/processed/	02_clean.py

Terminal

GitHub

Step 18 — Commit and push

```
git add code/03_descriptives.py
```

```
git add code/04_regression.py
```

```
git add README.md
```

```
git commit -m "session 6: R&D; intensity research design,
descriptives and regression"
```

```
git push
```

■ Do NOT add data/ or output/ — these are in .gitignore.

G Final checklist — tick everything before Session 7

■ README.md updated: RQ, H1, H2, variable table reflect R&D; intensity design

■ README.md note added explaining why DOI was dropped

■ 03_descriptives.py: data quality filters (at>0.1, sale>0, seq>0) present

■ 03_descriptives.py: at>=1 filter added to remove micro-firms

■ 03_descriptives.py: $Y = roa = ib/at$

■ 03_descriptives.py: $X = rd_intensity = xrd.fillna(0)/at$

■ 03_descriptives.py: $interaction = rd_x_size = rd_intensity * \ln_at$

■ 03_descriptives.py: CORE_VARS and WINSORIZE_VARS updated (no DOI)

■ 03_descriptives.py runs without errors

■ output/tables/summary_statistics.csv exists — values look reasonable

■ output/figures/correlation_matrix.png — no pair $|r| > 0.8$

- output/figures/main_relationship.png — no spike or wall of dots
- output/figures/dv_distribution.png — firm size axis starts above -2
- data/processed/panel_with_vars.parquet exists
- 04_regression.py: get_se() helper present (fixes AttributeError)
- 04_regression.py: DV, X_MAIN, INTERACT, CONTROLS updated
- 04_regression.py runs without errors — three models complete
- output/tables/regression_results.csv exists
- Results interpreted: H1 direction and significance noted
- Results interpreted: H2 interaction noted (significant or not)
- OLS vs FE difference noted — omitted variable bias discussed
- 2-3 plain-language bullet points written summarizing main findings
- task all runs end-to-end without errors
- .gitignore verified — data/ and output/ excluded
- 03_descriptives.py, 04_regression.py, README.md committed and pushed

Next session (Session 7 — Final): We write the research note in Quarto, insert tables and figures from output/, add BibTeX citations, render to PDF, and do a peer reproducibility check. Make sure task all runs clean and all output files exist before Session 7. Bring your 2-3 bullet point summary of results — you will need it for the write-up. Reference: github.com/vkiefner/sme-intl